Unit: Unit 0: Foundations of Algebra

Topic: GCF LCM

Name: ________________

Date: __________

Foundation (Jalapeno)

Q1. Find the GCF of 12 and 18 using factors.
- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 9

Q2. What is the LCM of 4 and 6 using multiples?
- (A) 4
- (B) 6
- (C) 8
- (D) 12
- (E) 24

Q3. Find the prime factorization of 45 to determine its GCF with 15.
- (A) $3 \cdot 5$
- (B) $3 \cdot 3 \cdot 5$
- (C) $3^2 \cdot 5$
- (D) $3 \cdot 5^2$
- (E) $3^2 \cdot 5^2$

Q4. A business has stocks priced at 12 and 18 per share. What is the LCM of these prices?
- (A) 12
- (B) 18
- (C) 36
- (D) 54
- (E) 108

Q5. Determine the GCF of 24 and 30 using prime factorization.
- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q6. A savings account earns 3% and 6% interest on two separate investments. Find the GCF of these interest rates.
- (A) 1%
- (B) 2%
- (C) 3%
- (D) 6%
- (E) 9%

Q7. What is the LCM of 8 and 10 using prime factorization?
- (A) 8
- (B) 10
- (C) 20
- (D) 40
- (E) 80

Q8. Determine the LCM of 9 and 12 to calculate the total amount of tax owed.
- (A) 9
- (B) 12
- (C) 18
- (D) 27
- (E) 36

Open Ended Questions (FRQ)

Q9. A company has two investments, one with a 5% annual return and another with a 10% annual return. Find the LCM of these interest rates and explain how it can be used to calculate the total return on investment over a 2-year period.

Q10. In a stock market, two stocks have prices of 15 and 25 per share. Find the GCF and LCM of these prices and discuss how they can be used to make informed investment decisions. Show your calculations and provide a brief explanation.

Chef's Tips

- Tip 1: Encourage students to use prime factorization to find GCF and LCM.
- Tip 2: Emphasize the importance of understanding the concept of GCF and LCM in real-world applications.
- Tip 3: Provide feedback on the free-response questions to help students improve their problem-solving skills.